

# DBM-SI / In-Situ Structural Intelligence

## Common Abbreviations and Terminology

### Introduction

This document provides a quick-reference glossary for common abbreviations, runtime concepts, and architectural terms used throughout:

- DBM-SI
- In-Situ Structural Intelligence
- PDS
- CGCCC
- ASB
- Structural Runtime Systems
- Robotics / Industrial AI integrations
- and related repositories.

The goal is not to enforce rigid terminology, but to help new readers rapidly navigate the structural intelligence ecosystem.

### Core Structural Intelligence Terms

| Abbreviation | Full Name                                     | Short Description                                                                                        |
|--------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------|
| DBM-SI       | Digital Brain Model – Structural Intelligence | Structural intelligence paradigm emphasizing IR, metric structures, CCC, dispatch, policy, and evolution |
| SI           | Structural Intelligence                       | Intelligence emerging from structural organization rather than purely monolithic transformation          |
| IR           | Intermediate Representation                   | Structural representation layer between raw input and intelligence/runtime operations                    |

|                |                                    |                                                                                              |
|----------------|------------------------------------|----------------------------------------------------------------------------------------------|
| CCC            | Common Concept Core                | Shared structural attractor/core across representations, behaviors, trajectories, or systems |
| Static CCC     | Static Common Concept Core         | Structural CCC independent of time/behavior                                                  |
| Behavioral CCC | Behavioral Common Concept Core     | CCC emerging through trajectory and behavior evolution                                       |
| HLM            | Hyper Language Model               | Structural extension beyond traditional LLM systems                                          |
| LLM            | Large Language Model               | Transformer-based large-scale semantic generative model                                      |
| VLM            | Vision Language Model              | Multimodal visual-language generative model                                                  |
| CAIG           | Component-Aware Intelligence Graph | Structural ecosystem of evolving intelligence components                                     |
| CGCCC          | Calling Graph Common Concept Core  | CCC extraction and governance framework over calling graphs                                  |
| SOS            | Start / Operation / State          | Structural encoding model used in CGCCC and runtime systems                                  |
| PDS            | Policy Decision System             | Structural decision/control runtime model                                                    |
| MES            | Manufacturing Execution System     | Operational industrial feedback/control layer                                                |
| MET            | Minimal Evolution Threshold        | Practical incremental evolution strategy emphasizing low-cost structural gains               |

|      |                                          |                                                                                  |
|------|------------------------------------------|----------------------------------------------------------------------------------|
| EDT  | Euclidean Differential Tree              | Differential-tree structure for spatial/metric partitioning and dispatch         |
| ASB  | Automatic Segmentation and Branching     | Structural dispatch/specialization paradigm using local branches or nodes        |
| TRI  | Trajectory Risk Intelligence             | Structural risk analysis framework using trajectory-aware intelligence           |
| SIEE | Structural Intelligence Evolution Engine | Runtime evolution layer for structural systems                                   |
| MG   | Motif Graph                              | Reusable motif-level graph abstraction used in structural bridging and evolution |

### PDS Runtime Terms

| Abbreviation                       | Full Name                             | Description                                           |
|------------------------------------|---------------------------------------|-------------------------------------------------------|
| S                                  | Segmentation / Structure Localization | Structural partitioning and localization layer        |
| K                                  | Knowledge Layer                       | Structural extraction and interpretation layer        |
| D                                  | Decision Layer                        | Decision-making runtime layer                         |
| P                                  | Policy Layer                          | Governance, safety, operational policy layer          |
| M                                  | Management / MES / Execution Layer    | Real-world execution and operational layer            |
| $X \rightarrow f(X) \rightarrow Y$ | Prediction Runtime Pattern            | Predictive runtime abstraction used in PDS            |
| PDS(u)                             | Policy Decision System Runtime        | Full runtime structural mapping for user/system state |

### Metric Space and Structural Geometry Terms

| Abbreviation | Full Name | Description |
|--------------|-----------|-------------|
|--------------|-----------|-------------|

|                     |                                 |                                                                    |
|---------------------|---------------------------------|--------------------------------------------------------------------|
| Metric Space        | Metric Space                    | Geometry defined through distance relationships                    |
| Euclidean Space     | Euclidean Space                 | Traditional geometric coordinate space                             |
| Metric Targeting    | Metric Targeting                | Navigation/search through structural distance relationships        |
| Metric Homing       | Metric Homing                   | Structural convergence toward CCC or target regions                |
| Structural Distance | Structural Distance             | Distance over structural relationships rather than raw coordinates |
| Directional Metric  | Directional Metric              | Asymmetric forward/backward structural distance                    |
| BTP                 | Bidirectional Traversal Pairing | Structural matching mechanism across graphs/paths                  |
| Unaligned AND       | Unaligned Structural Matching   | Structural pairing without strict alignment assumptions            |

## Structural Runtime and Evolution Terms

| Abbreviation        | Full Name                       | Description                                                             |
|---------------------|---------------------------------|-------------------------------------------------------------------------|
| In-Situ SI          | In-Situ Structural Intelligence | Structural intelligence formed directly within operational environments |
| Structural Runtime  | Structural Runtime System       | Runtime composed of evolving structural components                      |
| Component Formation | Structural Component Formation  | Creation of localized intelligence units                                |
| Structural Dispatch | Structural Dispatch             | Routing tasks to local structural components                            |

|                       |                                   |                                                   |
|-----------------------|-----------------------------------|---------------------------------------------------|
| Local Intelligence    | Localized Structural Intelligence | Specialized intelligence inside a local component |
| Structural Evolution  | Structural Component Evolution    | Split / merge / promotion / decay of components   |
| Evolution Layer       | Structural Evolution Layer        | Runtime evolution/ governance subsystem           |
| Structural Governance | Structural Governance             | Policy-aware control of structural evolution      |
| Evidence Chain        | Evidence Chain                    | Traceable validation and operational audit system |
| Quarantine Mode       | Quarantine Mode                   | Isolation mode for unstable or unsafe evolution   |
| Rollback              | Rollback Runtime                  | Reversion to prior structural/runtime state       |

## Robotics and Industrial AI Terms

| Abbreviation          | Full Name                                | Description                                                 |
|-----------------------|------------------------------------------|-------------------------------------------------------------|
| CPS                   | Cyber-Physical System                    | Integrated computational and physical system                |
| PLC                   | Programmable Logic Controller            | Industrial control runtime                                  |
| SCADA                 | Supervisory Control and Data Acquisition | Industrial supervisory control infrastructure               |
| MES Feedback          | MES Feedback Loop                        | Operational feedback driving runtime adaptation             |
| Structural Controller | Structural AI Controller                 | Componentized AI-driven industrial controller               |
| Robot Runtime         | Structural Robotics Runtime              | Robotics intelligence operating through evolving components |

## CGCCC / AI Coding Terms

| Abbreviation          | Full Name                         | Description                                                    |
|-----------------------|-----------------------------------|----------------------------------------------------------------|
| CG                    | Calling Graph                     | Structural graph representing software execution relationships |
| CG_high               | High-Level Calling Graph          | High-level architectural graph abstraction                     |
| CG_min                | Minimal Granularity Calling Graph | Statement-level or minimal operation graph                     |
| Vertical Gap Bridging | Vertical Structural Bridging      | Bridging architecture-level intent to implementation details   |
| ODGZ                  | On-Demand Granularity Zooming     | Dynamic structural zooming for AI coding/runtime analysis      |
| Compliance Gauge      | Structural Compliance Gauge       | Structural evaluation system for AI-generated code             |
| Rewrite Gradient      | Structural Rewrite Gradient       | Metric-driven structural refinement process                    |

## Intelligence and Evolution Concepts

| Abbreviation         | Full Name                                                                 | Description                                       |
|----------------------|---------------------------------------------------------------------------|---------------------------------------------------|
| APTGOE               | Autonomy, Parameterization, Training, Generation, Optimization, Evolution | Structural intelligence evolution ladder          |
| Structural Attractor | Structural Attractor                                                      | Stable convergence region in structural evolution |
| CCC Attractor Field  | CCC Attractor Field                                                       | Structural convergence field around CCC regions   |
| Stability Frontier   | Structural Stability Frontier                                             | Boundary between stable and unstable evolution    |
| Evolutionary Runtime | Evolutionary Runtime System                                               | Runtime capable of self-refinement and adaptation |

|                      |                                   |                                                          |
|----------------------|-----------------------------------|----------------------------------------------------------|
| Structural Ecosystem | Structural Intelligence Ecosystem | Collection of interacting evolving structural components |
|----------------------|-----------------------------------|----------------------------------------------------------|

## High-Level Conceptual Summary Traditional AI

Input

→ Universal Model

→ Output

## Structural Intelligence

World IR



Structural Components



Metric Dispatch



Localized Intelligence



Policy / Feedback



Structural Evolution

## One-Line Orientation

DBM-SI and In-Situ Structural Intelligence investigate how intelligence can emerge from evolving structural systems operating over IR, metric spaces, feedback loops, policies, and localized component ecosystems.